

LESSON 7-4 • HOMEWORK PACKET • SHEET 1 OF 2**Comparing Numbers on a Number Line**

Maryland Standard 6.NS.C.7a • Learning target – I can compare two rational numbers using their positions on a number line.

REMEMBER

- Numbers get **greater to the right** and **less to the left**.
- Any positive number is greater than any negative number.
- $-2 > -8$ because -2 is farther right. $<$ means is less than; $>$ means is greater than.

WORKED EXAMPLE

Compare -3.5 and -1.5 . Use $<$ or $>$.

Step 1 -1.5 is to the right of -3.5 on the number line.

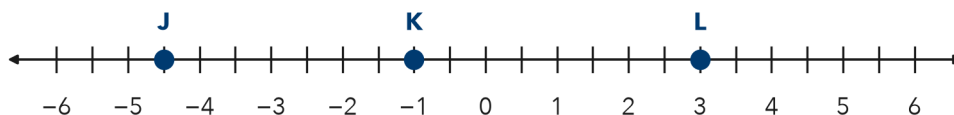
Step 2 Right means greater.

Answer $-1.5 > -3.5$

PRACTICE

show your work · label units

1. Use the number line. Write $<$ or $>$ in each blank.



a. J ____ K

b. L ____ K

c. J ____ L

2. Write $<$ or $>$ in each blank.

a. -8 ____ -3

b. 0 ____ -6

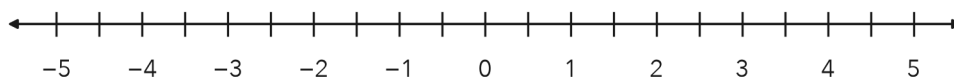
c. -2.5 ____ -2.05

d. 4 ____ -10

e. $-1/2$ ____ $-3/4$

f. -7 ____ -7.1

3. Plot -3.5 and -0.5 on the number line. Then write a true comparison using $<$. _____



4. **Golf.** In golf, the lower score wins. Tanya scored -3 . Malik scored -5 .

a. Compare the scores using $<$ or $>$. _____

b. Who won? _____

5. **Cold Days.** Monday's low was -7 degrees. Tuesday's low was -2 degrees. Which day was colder? Write a comparison using $<$ or $>$.

Answer _____

6. **Reasoning.** Is $-6 < |-6|$ true? Explain how you know.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Ask someone at home: which is colder, -4 degrees or -9 degrees? Why?

LESSON 7-4 • HOMEWORK PACKET • SHEET 2 OF 2

Ordering Integers and Rational Numbers

Maryland Standard 6.NS.C.7b • Learning target – I can order integers and rational numbers from least to greatest.

REMEMBER

- Place the numbers on a number line. **Least** is farthest left; **greatest** is farthest right.
- Change fractions to decimals to compare: $-3/4 = -0.75$.
- For negative numbers, the one farther from 0 is **less**.

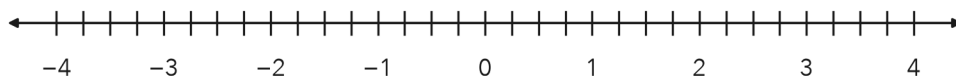
WORKED EXAMPLE

Order -1.2 , 0.5 , -2 , and $-1/2$ from least to greatest.**Step 1** Write $-1/2$ as -0.5 .**Step 2** Read the number line from left to right.**Answer** -2 , -1.2 , $-1/2$, 0.5

PRACTICE

show your work • label units

1. Plot each number. Then write them in order from least to greatest:
- 1.5
- ,
- -2.5
- ,
- $-3/4$
- ,
- $2\frac{1}{4}$
- ,
- -3.25



Order _____

2. Order each set from least to greatest.

a. -6 , 4 , -9 , 0 _____b. -0.3 , -3 , -0.03 _____

3. Circle the greatest number in each set.

a. -5 , -2 , -8 b. -1.4 , -1.04 , -1.44 c. $-2/3$, $-1/2$, $-3/4$ d. 0 , -0.1 , -10

- 4.
- Ocean Animals.**
- A shark swims at
- -45.5
- m, a turtle at
- -12
- m, a dolphin at
- -30.25
- m, and a jellyfish at
- -8.75
- m. List the animals from deepest to closest to the surface.

Answer _____

- 5.
- Find the Error.**
- Leo ordered
- $-1/2$
- ,
- -0.4
- , and
- -0.6
- from least to greatest as
- -0.4
- ,
- $-1/2$
- ,
- -0.6
- .

Explain his mistake and write the correct order.

TEST PRACTICE

choose the best answer

6. Which list is ordered from least to greatest?

A -5 , -8 , 0 , 3 B -8 , -5 , 0 , 3 C 0 , -5 , -8 , 3 D 3 , 0 , -5 , -8

7. Which number is greatest?

A -2.7 B -2.07 C -2.77 D -2.2

CHECK YOURSELF

☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** List the low temperatures for three cold days with someone at home. Order them from coldest to warmest.